

Introductory Econometrics with Recitation: TA Session Week 13

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Today's agenda

1 Unique Function

- `unique()`

2 Panel Data

- Brief View
- Model Specification
- Model Estimation
- Inference in Panel Data

Today's Dataset

- Today we will use the following CSV file:¹
 - `guns.csv`: a panel dataset on U.S. states from 1977 to 1999.
-

```
gun <- read.csv("data/guns.csv")  
  
gun <- gun %>%  
  mutate(lnvio = log(vio))
```

¹You can download the dataset [here](#).

Extract distinct values: unique()

- unique() keeps only the distinct values in a vector.
- **Example:**

```
fruit <- c("apple", "banana", "apple", "orange", "banana")  
  
unique(fruit)
```

- **Output:**

```
[1] "apple" "banana" "orange"
```

- It is useful when we want to know what values appear in a variable.

A Brief View of Panel Data

Variable	Description
stateid	State ID
year	Year
lnvio	Log of violent crime rate
shall	= 1 if state has shall-carry law; 0 otherwise
incarc_rate	Incarceration rate last year
density	Population per 1000 square miles
avginc	Real per capita income
pop	State population

A Brief View of Panel Data

- Data:

stateid	year	vio	lnvio	...
1	77	414.4	6.026832	...
1	78	419.1	6.038110	...
1	79	413.3	6.024174	...
1	80	448.5	6.105909	...
1	81	470.5	6.153796	...
1	82	447.7	6.104123	...

- In panel data, we observe the same unit repeatedly over time.
- In this dataset, the unit is a state and the time period is a year.
- Together, stateid and year identify each observation.

Story: Carrying Guns and Crime?

- Some U.S. states have enacted laws allowing citizens to carry concealed weapons.
- Do these laws affect the crime rate?
 - Crime may decrease if criminals are deterred.
 - Crime may increase if more people carry guns.
- To study the effect, we need to account for state heterogeneity and time trends.
 - Some states have higher crime rates than others.
 - Crime rates may change over time.

Model Specification

- We begin with the simple model:

$$\ln(vio_{it}) = \beta_0 + \beta_1 shall_{it} + \varepsilon_{it}.$$

- Then we add control variables:

$$\ln(vio_{it}) = \beta_0 + \beta_1 shall_{it} + \beta_2 incarc_rate_{it} + \cdots + \varepsilon_{it}.$$

- The coefficient of interest is β_1 .

Model Specification

- Next, we add state fixed effects:

$$\ln(vio_{it}) = \alpha_i + \beta_1 shall_{it} + \beta_2 incarc_rate_{it} + \cdots + \varepsilon_{it}.$$

- Finally, we add both state and year fixed effects:

$$\ln(vio_{it}) = \alpha_i + \lambda_t + \beta_1 shall_{it} + \beta_2 incarc_rate_{it} + \cdots + \varepsilon_{it}.$$

- We can estimate fixed-effect models using `feols()` from the `fixest` package.

Model Estimation: Models (1) and (2)

• Model (1): Simple OLS model

```
model_1 <- lm_robust(  
  lnvio ~ shall,  
  data = gun  
)
```

• Model (2): OLS model with controls

```
model_2 <- lm_robust(  
  lnvio ~ shall + density + avginc + pop + pb1064 +  
    pw1064 + pm1029,  
  data = gun  
)
```

Model Estimation: Model (3)

- **Model (3): One-way fixed effect model**
- We use | stateid to add state fixed effects.

```
library(fixest)

model_3 <- feols(
  lnvio ~ shall + incarc_rate + density + avginc +
    pop + pb1064 + pw1064 + pm1029 | stateid,
  data = gun
)
```

- The | operator separates regressors from fixed effects.

Model Estimation: Model (4)

- **Model (4): Two-way fixed effect model**
- We use `| stateid + year` to add state and year fixed effects.

```
model_4 <- feols(  
  lnvio ~ shall + incarc_rate + density + avginc +  
    pop + pb1064 + pw1064 + pm1029 | stateid + year,  
  data = gun  
)
```

Estimation Results

Output:

	Simple	Control	OWFE	TWFE
(Intercept)	6.13 *** (0.02)	4.20 *** (0.66)		
shall	-0.44 *** (0.05)	-0.34 *** (0.04)	-0.05 (0.04)	-0.03 (0.04)
density		0.10 *** (0.02)	-0.17 (0.14)	-0.09 (0.12)
avginc		0.00 (0.01)	-0.01 (0.01)	0.00 (0.02)
pop		0.05 *** (0.00)	0.01 (0.01)	-0.00 (0.02)
pb1064		0.10 *** (0.02)	0.10 ** (0.03)	0.03 (0.05)
pw1064		0.03 ** (0.01)	0.04 ** (0.01)	0.01 (0.02)
pm1029		-0.06 *** (0.01)	-0.05 * (0.02)	0.07 (0.05)
incarc_rate			-0.00 (0.00)	0.00 (0.00)
R ²	0.09	0.48		
Adj. R ²	0.09	0.48		
Num. obs.	1173	1173	1173	1173
RMSE	0.62	0.47		
Num. groups: stateid			51	51
R ² (full model)			0.94	0.96
R ² (proj model)			0.22	0.06
Adj. R ² (full model)			0.94	0.95
Adj. R ² (proj model)			0.21	0.05
Num. groups: year				23

*** p < 0.001; ** p < 0.01; * p < 0.05

Inference in Panel Data

- Do not forget to adjust standard errors under the panel data structure.
- Unlike cross-sectional data, errors may be correlated across time within the same unit.
- For example, we can cluster standard errors by state.
 - Errors are allowed to be correlated within each state.
 - Errors are assumed to be independent across states.
- In `fixest`, use `summary()` and specify the cluster variable through `vcov`.

```
summary(model_4, vcov = "iid") # assume homoskedasticity  
summary(model_4, vcov = ~ stateid) # cluster by stateid
```

- By default, `fixest` may already use clustered standard errors when fixed effects are included.

Inference in Panel Data

• Output:

```

OLS estimation, Dep. Var.: lnvio
Observations: 1,173
Fixed-effects: stateid: 51, year: 23
Standard-errors: Clustered (stateid)

              Estimate Std. Error   t value Pr(>|t|)
shall        -0.027994   0.040717  -0.687518  0.49493
incarc_rate   0.000076   0.000208   0.365556  0.71624
density      -0.091555   0.123862  -0.739167  0.46326
avginc        0.000959   0.016493   0.058124  0.95388
pop          -0.004754   0.015229  -0.312188  0.75620
pb1064        0.029186   0.049541   0.589135  0.55842
pw1064        0.009250   0.023756   0.389374  0.69865
pm1029        0.073325   0.052473   1.397384  0.16847
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.135105      Adj. R2: 0.952971
                  Within R2: 0.056351

```